Leonid Sokolov

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EDUCATION

UCL, *MSc Data Science and Machine Learning* **Sep 2026 – Sep 2027 (expected)**

- Selected modules: Supervised Learning, Applied Machine Learning, Numerical Optimisation, Bayesian Deep Learning.

University of Manchester, *BEng (Hons) Electronic Engineering* **Sep 2023 – Jun 2026**

- First class (72%); dissertation: 74%; Mathematics 2E1 (88%), C Programming (85%), Concurrent Systems (76%).

Concord College, *Shrewsbury* **Sep 2021 – Jun 2023**

- A Levels: Mathematics A*, Further Mathematics A, Physics A.

PROJECT EXPERIENCE

Enhava.app *(ML video-upscaling service, in development)* **Jun 2026 – Present**

- Built a Python service around pretrained SeedVR2, orchestrating on-demand Vast.ai GPU workers for video upscaling.
- Designed a distributed video-processing workflow to handle worker and network failures using heartbeats, bounded retries and checkpoint recovery, with output validation before publication.
- Built the TypeScript/React frontend through Codex agent pipelines I designed, covering uploads, processing quotes, progress and downloads.

Dissertation: Neural Networks and Model Predictive Control **2025 – 2026**

- Implemented neural-network policy training from scratch in Python and NumPy, with backpropagation through simulated dynamics and Adam optimisation; compared controllers across nine first-order systems and a cart-pole.
- Built a configurable experiment pipeline: a shared manifest defines controller variants, network architectures and test scenarios for concurrent training and evaluation, with saved weights reused across runs.

Line-following buggy *(University of Manchester)* **Second-year group project**

- Designed and tuned the team's C++ PID line-following controller; the buggy placed 6th of 50 by speed, completing its run without line-following errors.

EXPERIENCE

DomiDO

Software Developer **May 2026 – Present**

- Designed and used a Codex agent pipeline to coordinate feature implementation from detailed specifications, with reviewers and a judge checking requirements and test evidence.
- Built GitLab CI/CD pipelines for a Go backend and frontend; independent pipelines run tests, builds and previews, then publish images for deployment to K3s through a separate infrastructure repository.

VirtuosoHub *(videoconferencing for musicians)*

Project Manager (backend engineering and DevOps) **Dec 2024 – May 2026**

- Refined transcript collection in existing LiveKit AI agents, evaluated alternative transcription models on a test bench and integrated the selected model.
- Designed prompts and structured outputs to turn lesson transcripts into a dynamic feedback interface highlighting learners' strengths, areas for improvement and practice priorities.
- Migrated a WebRTC signalling prototype to LiveKit; supported three LiveKit agent nodes and two backend nodes on Hetzner using K3s and Grafana.

Orb *(social media app)*

Co-Founder & Software Engineer **Jul 2023 – Nov 2024**

- Built JWT-protected Python APIs and fuzzy-logic location processing on GCP/Firebase; coordinated a three-person team.

A-Exam *(chemistry revision app)*

Founder and Software Engineer **2019 – 2020**

- Reached 7,500+ Google Play downloads by building a Java chemistry-revision app with topic-level progress tracking.

SKILLS

- Programming: Python (NumPy), C++, Go, Java; C (coursework); PyTorch (learning).
- Infrastructure: Linux (basic command-line use), Git, GitLab CI/CD, K3s/Kubernetes, LiveKit, Vast.ai, GCP/Firebase, AWS S3, Grafana.